Hadoop & Spark - ISEN 2025/2026

It's assignment time !

Exercice presentation

The goal of this assignment is to present a report of your choice in a Colab Notebook using Spark Dataframes.

You are free to choose any Kaggle dataset(s) of your choice but you need to use the following command :

```
!kaggle datasets download -d user/dataset-name
```

The report could be about any theme you like (execept the examples seen in this course), for example a study of the price of the houses by area, the average delay in public transportations, studying the sales of a company ...

Originality is part of the notation ! 🚩✔️

Rules

You are free to do as you want as long as you follow those rules:

- When I run your notebook for evaluation, **I should be able to launch every cells without any intervention from my part** (except for the Kaggle and Ngrok credentials)
- Start from the template `Colab_starting_template.ipynb` and do not change the dependencies part (you may add new ones but not delete them).
- You are using only PySpark to handle the data (**no Pandas or any similar tool**) with Dataframes and RDDs only.
    - The only exception to that is if you want to use libraries such as `matplotlib` that uses pandas objects, you can use the `.toPandas()` method but you are not allowed to change anything about the data in Pandas (do it in Spark before converting).

Rules

- You should comment every cell and explain what you do and your thought process. Show me you understand the commands you are using and do not hesitate to cite the course (show me you are not Chat GPT! 🤖)

For example:

I now aggregate the data by `PickupLocationId` and show the average trip time in minute.

```
[135] yellowTaxiReport = (
          yellowTaxiDF
            .groupBy("PickupLocationId")
            .agg(
                avg("TripTimeInMinutes").alias("AvgTripDuration")
            )
            .orderBy(col("PickupLocationId").desc())
      )
      yellowTaxiReport.show()
```

- Feel free to add schemas, graphs and pictures but make sure I can see them once you send me you notebook.

Rules

- You will need to include:
  - filters
  - data cleaning
  - calculated columns
  - convert time format
  - aggregations
  - Merges, joins, unions or similar operations between 2 dataframes
- I want you to use at least two functions or libraries not seen in this course (check the doc for ideas !), show me you are a capable Engineer 😎

  But do not limit yourself to that !

Rules

- You need to **declare your schema manually** (no infer schema)
- Show me at least one screenshot of the Spark UI and explain me the showed plan (transformations, actions ...)
- Put your name at the beginning of the Notebook **AND** in the file name (for example `Project_Alexandre_Verept.ipynb`)

Rules

- Obviously I do not want to see the exact same Dataframes and ideas on two different student submissions, the internet is vast and open, you can find your own examples and not copy your neighbour 👍

Example of matplotlib usage (not mandatory)

Handle your data with PySpark

```
data = [("A", 10), ("B", 20), ("C", 30), ("D", 40)]
columns = ["Category", "Value"]
rows = [Row(*item) for item in data]
df = spark.createDataFrame(rows, columns)
```

Graph plot using to pandas (no manipulation of data allowed with Pandas):

```
import matplotlib.pyplot as plt

pdf = df.toPandas()
plt.bar(pdf["Category"], pdf["Value"])
plt.xlabel("Category")
plt.ylabel("Value")
plt.title("Bar Plot Example")
plt.show()
```

Due date

Due date: **27th of February 2025**

Send your notebook as `.ipynb` (and screenshots if needed) at `alexandre.verept.tech@outlook.com`

Due date

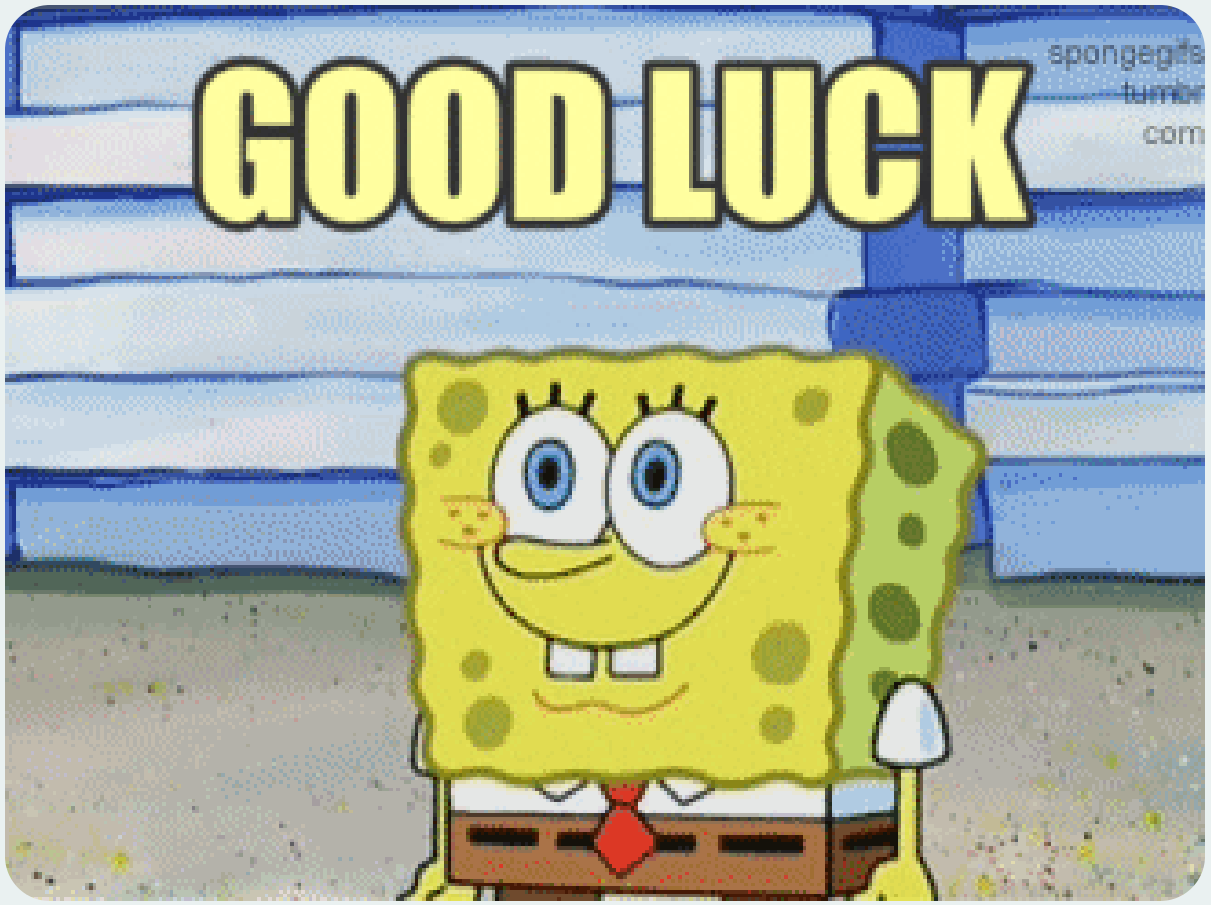
I may slightly change the notations to adapt to your submissions, but this is a general idea:

Criteria	Points
Originality of the report theme	2
Properly commented cells and explanations	5
Manual schema declaration	1
Usage of filter	1
Data cleaning	2
Calculated columns	2
Time format conversion	1
Aggregation	2
Merge, join, union, or similar operations between 2 dataframes	2
Screenshot and explanation of the Spark UI	2
Total	**20**

FAQ

- Can I sent a report in Chinese or Spanish ?
  - It should be either French or English
- Can I send it 3 weeks after the due date ?
  - 25th february 2025 at 23:59
- Can we make groups of two or more ?
  - This is a personnal assignment
- Can I send it through a Github link ?
  - Preferrably not, sent it directly in the email
- Can I send an anonymous work ?
  - If you want me to be able to give a note, please put your name in the filename + in the header + in the mail

Good luck !



Credit and used resources:

- Hadoop & Spark courses from previous years by Olivier Moïse (2022) & Fabrice Woittequand (2020)